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**https://github.com/jesus-garcia-izquierdo/N-body-problem**

# The gravitational $N$-body problem

Symplectic integration, conservation laws, stability and chaos

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Python code, videos and data: github.com/jesus-garcia-izquierdo/N-body-problem

**Abstract.** We study the motion of $N$ point masses interacting only through Newtonian gravity. After reviewing the physics (equations of motion, Hamiltonian structure, conservation laws, the virial theorem, linear stability and Lyapunov exponents) we integrate the equations numerically with the velocity Verlet method and four alternative schemes, using a fully vectorized Python implementation. Eight experiments are analysed: the Sun–Earth–Moon system, a four-body system with a massive central body, a chaotic six-body system, a comparison of integrators, the figure-eight choreography, the stability of Lagrange's equilateral triangle, the Solar System and the gravitational collapse of a star cluster. Energy is conserved to $6.41 \times 10^{-10}$ (Sun–Earth–Moon, two years), linear and angular momentum to machine precision, the measured orders of convergence match theory, the instability rate of Lagrange's triangle agrees with linear theory to three digits (1.225 vs. 1.225), Kepler's third law is recovered with slope 1.4989, and the six-body system has a Lyapunov time of only 108 days.

This report extends and corrects the 2023 project *Problema de los n-cuerpos* by G. Haro Pérez and J. García Izquierdo (MATLAB implementation).

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# 1 Introduction

The $N$-body problem asks for the motion of $N$ masses that attract each other gravitationally. For $N = 2$ it was solved by Newton: the bodies move on conic sections. For $N \geq 3$ no general closed-form solution exists. Bruns (1887) showed that there are no further algebraic integrals beyond the ten classical ones, and Poincaré (1890) showed that the motion can be extremely sensitive to the initial conditions, the phenomenon we now call *chaos*. Sundman (1912) did find a convergent power series for $N = 3$, but it converges far too slowly to be of any practical use.

The only general way forward is therefore numerical integration. A numerical solution, however, is only trustworthy if it respects what the exact dynamics guarantees: conservation of energy, momentum and angular momentum, and statistical relations such as the virial theorem. This report has three goals:

1. to present the physics needed to interpret a simulation (Section 2);
2. to explain why a *symplectic* integrator such as velocity Verlet is the right tool, and to quantify its accuracy (Section 3);
3. to analyse eight experiments that range from the Solar System to chaos and star-cluster formation (Section 4).

Every number quoted in the text is produced by the code in the repository; the script `tools/build_documents.py` regenerates all figures and values.

# 2 Physical model

## 2.1 Equations of motion

Let body $i$ have mass $m_i$, position $\mathbf{r}_i(t)$ and velocity $\mathbf{v}_i(t)$. Newton's law of gravitation together with the superposition principle gives

$$\ddot{\mathbf{r}}_i = G \sum_{j \neq i} m_j \frac{\mathbf{r}_j - \mathbf{r}_i}{|\mathbf{r}_j - \mathbf{r}_i|^3}, \qquad i = 1, \ldots, N, \tag{1}$$

with $G = 6.67430 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$. These are $3N$ coupled second-order ordinary differential equations, or $6N$ first-order ones for $(\mathbf{r}_i, \mathbf{v}_i)$. Given positions and velocities at $t = 0$ the solution exists and is unique as long as no two bodies collide.

## 2.2 Hamiltonian structure

Writing the energy in terms of positions $\mathbf{q}_i = \mathbf{r}_i$ and momenta $\mathbf{p}_i = m_i \mathbf{v}_i$,

$$H(\mathbf{q}, \mathbf{p}) = \underbrace{\sum_i \frac{|\mathbf{p}_i|^2}{2m_i}}_{T(\mathbf{p})} \underbrace{- \sum_{i<j} \frac{G m_i m_j}{|\mathbf{q}_i - \mathbf{q}_j|}}_{V(\mathbf{q})}, \tag{2}$$

Hamilton's equations $\dot{\mathbf{q}}_i = \partial H / \partial \mathbf{p}_i$, $\dot{\mathbf{p}}_i = -\partial H / \partial \mathbf{q}_i$ reproduce Eq. (1). Two properties of the exact flow will guide the choice of integrator: it preserves volume (indeed the symplectic two-form $\sum_i \mathrm{d}\mathbf{q}_i \wedge \mathrm{d}\mathbf{p}_i$) in the $6N$-dimensional phase space (Liouville's theorem), and it is time-reversible. Note also that $H = T(\mathbf{p}) + V(\mathbf{q})$ is *separable*, which is what makes the Verlet splitting possible (Section 3.1).

### 2.3 Conservation laws

By Noether's theorem, the symmetries of an isolated system imply conserved quantities:

$$\text{translations} \;\Rightarrow\; \mathbf{P} = \sum_i m_i \mathbf{v}_i, \tag{3}$$

$$\text{rotations} \;\Rightarrow\; \mathbf{L} = \sum_i \mathbf{r}_i \times m_i \mathbf{v}_i, \tag{4}$$

$$\text{time shifts} \;\Rightarrow\; E = K + U, \quad K = \tfrac{1}{2}\sum_i m_i v_i^2, \quad U = -\sum_{i<j} \frac{G m_i m_j}{r_{ij}}. \tag{5}$$

The proofs show what a numerical method must respect. $\dot{\mathbf{P}} = 0$ follows from $\mathbf{F}_{ij} = -\mathbf{F}_{ji}$ (Newton's third law), so any scheme built on exactly antisymmetric pair forces conserves $\mathbf{P}$ up to round-off. $\dot{\mathbf{L}} = 0$ follows because the forces are central, $\mathbf{F}_{ij} \parallel \mathbf{r}_i - \mathbf{r}_j$. The potential energy counts every pair once; summing over all ordered pairs and dividing by two, as in the original report, is equivalent.

Since $\mathbf{P}$ is constant, the centre of mass (CM) moves uniformly with $\mathbf{V}_{\rm cm} = \mathbf{P}/M$. We work in the CM frame ($\mathbf{P} = 0$) whenever possible: the system does not drift away, and the kinetic energy loses the trivial term $\frac{1}{2}MV_{\rm cm}^2$, which is essential for the virial theorem.

### 2.4 The virial theorem

Define the virial $\mathcal{G} = \sum_i \mathbf{p}_i \cdot \mathbf{r}_i$. Its derivative is $\dot{\mathcal{G}} = 2K + \sum_i \mathbf{F}_i \cdot \mathbf{r}_i$. Because $U$ is homogeneous of degree $-1$ in the positions, Euler's theorem gives $\sum_i \mathbf{r}_i \cdot \nabla_i U = -U$, i.e. $\sum_i \mathbf{F}_i \cdot \mathbf{r}_i = U$. Averaging over a time $\tau$,

$$\frac{\mathcal{G}(\tau) - \mathcal{G}(0)}{\tau} = 2\langle K\rangle + \langle U\rangle \xrightarrow[\tau\to\infty]{} 0 \quad\Longrightarrow\quad Q \equiv \frac{2\langle K\rangle}{-\langle U\rangle} = 1, \tag{6}$$

provided the motion is bounded. Three conditions are therefore required: a bound system ($E < 0$) with no escapers, $K$ measured in the CM frame, and a long averaging time. In equilibrium $E = \frac{1}{2}\langle U\rangle = -\langle K\rangle$.

### 2.5 The two-body problem

With $\mathbf{R}$ the CM and $\mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1$, the two-body problem reduces to one body of reduced mass $\mu = m_1 m_2/(m_1 + m_2)$ in a central field. The orbit is a conic; for bound orbits two results are used repeatedly below:

$$v^2 = GM\left(\frac{2}{r} - \frac{1}{a}\right) \text{ (vis-viva)}, \qquad T^2 = \frac{4\pi^2}{GM}\, a^3 \text{ (Kepler's third law)}, \tag{7}$$

where $a$ is the semi-major axis. A circular orbit has $v_c = \sqrt{GM/r}$.

### 2.6 Stability, chaos and Lyapunov exponents

Let $\delta(t)$ be the phase-space separation between two solutions that start a distance $\delta_0$ apart. For almost every initial direction

$$\lambda_{\max} = \lim_{t\to\infty} \lim_{\delta_0\to 0} \frac{1}{t} \ln \frac{|\delta(t)|}{\delta_0} \tag{8}$$

exists and is the *maximal Lyapunov exponent*. Regular motion has $\lambda_{\max} = 0$ (separations grow at most polynomially); chaotic motion has $\lambda_{\max} > 0$, and errors grow by a factor $e$ every Lyapunov time $t_L = 1/\lambda_{\max}$. With an initial uncertainty $\delta_0$ and a tolerance $\Delta$, predictions are possible only

up to $t_{\rm pred} \approx t_L \ln(\Delta/\delta_0)$: improving the measurements a thousandfold buys only $\ln 1000 \approx 7$ extra Lyapunov times.

We estimate $\lambda_{\max}$ with Benettin's method: a "shadow" trajectory is integrated alongside the reference one; every $\tau$ its separation $d_k$ is measured, $\ln(d_k/\delta_0)$ is accumulated and the shadow is rescaled back to distance $\delta_0$ along the same direction, so that $\lambda \approx t^{-1}\sum_k \ln(d_k/\delta_0)$. Positions and velocities are made dimensionless with the rms radius $L$ and the time $L/v_{\rm rms}$.

*Linear stability* is the local version of the same idea: a solution is linearly unstable if small perturbations grow exponentially. Lagrange's equilateral solution (Section 4.6) is a textbook example where the growth rate can be computed analytically and compared with the simulation.

### 2.7 Softening and units

For star clusters, where particles represent many stars and close two-body encounters are not of interest, we replace $1/r \to 1/\sqrt{r^2+\varepsilon^2}$ (Plummer softening). The force remains the gradient of the softened potential, so the softened energy is still conserved. Abstract problems use $N$-body units $G = M = 1$, in which all quantities are of order one; physical scales are recovered by noting that time scales as $\sqrt{L^3/GM}$.

## 3 Numerical method

### 3.1 Velocity Verlet

Writing $\mathbf{a}_n = \mathbf{a}(\mathbf{r}_n)$ for the accelerations of Eq. (1), one step of length $h$ is

$$\mathbf{r}_{n+1} = \mathbf{r}_n + h\,\mathbf{v}_n + \tfrac{h^2}{2}\,\mathbf{a}_n, \tag{9}$$

$$\mathbf{a}_{n+1} = \mathbf{a}(\mathbf{r}_{n+1}), \tag{10}$$

$$\mathbf{v}_{n+1} = \mathbf{v}_n + \tfrac{h}{2}\,(\mathbf{a}_n + \mathbf{a}_{n+1})\,. \tag{11}$$

The position update is a second-order Taylor expansion and the velocity update is the trapezoidal rule for $\int \mathbf{a}\,\mathrm{d}t$. Introducing $\mathbf{v}_{n+1/2} = \mathbf{v}_n + \frac{h}{2}\mathbf{a}_n$ the same algorithm reads "half kick – drift – half kick". This is the operator splitting $e^{\frac{h}{2}V}e^{hT}e^{\frac{h}{2}V}$ of the Hamiltonian (2): the drift is the exact flow of $T$ and the kicks are the exact flow of $V$. Hence

- the method is **symplectic** (a composition of Hamiltonian flows) and **time-reversible** (the composition is symmetric);
- by the Baker–Campbell–Hausdorff formula the local error is $\mathcal{O}(h^3)$ and the global error $\mathcal{O}(h^2)$;
- it conserves $\mathbf{P}$ and $\mathbf{L}$ exactly (each sub-step does);
- it needs one force evaluation per step, because $\mathbf{a}_{n+1}$ is reused as the $\mathbf{a}_n$ of the next step (the original MATLAB code evaluated it twice).

**Why symplectic matters.** Backward error analysis shows that a symplectic method of order $p$ follows *exactly* the dynamics of a modified, "shadow" Hamiltonian $\tilde{H} = H + h^p H_p + \dots$, up to exponentially small terms. The numerical trajectory conserves $\tilde{H}$, so the error in $H$ stays bounded by $\mathcal{O}(h^p)$ over exponentially long times instead of drifting. This is the single most important property for long orbital integrations.

### 3.2 Other integrators

To show the effect of order and structure, four other schemes are implemented (Table 1). Explicit Euler expands phase-space areas by $1+\omega^2h^2$ per step on a harmonic oscillator, so the energy grows exponentially; swapping two lines (symplectic Euler) makes the area factor exactly one. Yoshida's method composes three Verlet steps of lengths $w_1h$, $w_0h$, $w_1h$ with $w_1 = 1/(2-2^{1/3})$

and $w_0 = -2^{1/3}/(2 - 2^{1/3})$ (note the step backwards in time), which cancels the $h^3$ error term. RK4 is fourth order but not symplectic.

**Table 1:** Integrators implemented in `nbody/integrators.py`.

| Method | Order | Symplectic | Reversible | Forces per step |
|---|---|---|---|---|
| Explicit Euler | 1 | no | no | 1 |
| Symplectic Euler | 1 | yes | no | 1 |
| Velocity Verlet | 2 | yes | yes | 1 |
| Yoshida-4 | 4 | yes | yes | 3 |
| Runge–Kutta 4 | 4 | no | no | 4 |

### 3.3 Implementation

The code is a small Python package (`nbody`) built on NumPy. The state is stored as arrays of shape $(N, 3)$ (body $\times$ component) and a trajectory as arrays of shape $(T, N, 3)$ (time $\times$ body $\times$ component). Accelerations are computed without any Python loop over bodies: broadcasting builds the $(N, N, 3)$ tensor of separations $\mathbf{r}_j - \mathbf{r}_i$, the diagonal is excluded by setting $r_{ii}^2 = \infty$, and an Einstein summation contracts the index $j$. The cost is $\mathcal{O}(N^2)$ per step, as for any direct-summation code. Memory for the trajectory is allocated once (the MATLAB version grew its arrays at every step, an $\mathcal{O}(T^2)$ cost). Typical run times on a laptop are a second for the Sun–Earth–Moon system ($6.3 \times 10^4$ steps) and a few seconds for the Solar System ($1.2 \times 10^5$ steps).

### 3.4 Diagnostics and choice of time step

Conservation is measured *relative to the initial value*: $|E(t) - E_0|/|E_0|$ for the energy, and for the vectors $\mathbf{P}$ and $\mathbf{L}$ the norm of the change divided by the natural scales $\max_t \sum_i m_i|\mathbf{v}_i|$ and $\max_t \sum_i m_i|\mathbf{r}_i||\mathbf{v}_i|$ (needed because $\mathbf{P} = 0$ in the CM frame). The time step is set by the shortest dynamical time of the system, with the rule of thumb $h \lesssim t_{\mathrm{dyn}}/100$, and every result was checked by halving $h$.

## 4 Results

### 4.1 The Sun–Earth–Moon system

The bodies start in the ecliptic plane with the Earth at aphelion and the Moon at apogee (Table 2). The shortest period is the Moon's, 27.3 days, so $h = 1000\,\mathrm{s}$ gives about 2400 steps per lunar orbit.

**Table 2:** Initial conditions of the Sun–Earth–Moon system (before moving to the CM frame). Integration time 2 years, $h = 1000\,\mathrm{s}$.

| | Sun | Earth | Moon |
|---|---|---|---|
| Mass (kg) | $1.98847 \times 10^{30}$ | $5.9722 \times 10^{24}$ | $7.342 \times 10^{22}$ |
| $x_0$ (m) | 0 | $1.52098232 \times 10^{11}$ (aphelion) | $x_\oplus + 4.055 \times 10^8$ (apogee) |
| $v_{y,0}$ (m/s) | 0 | 29 290 | $v_\oplus + 964$ |

Figure 1 shows the orbits. At heliocentric scale the Moon's orbit is invisible (the Earth–Moon distance is 1/375 of the Sun–Earth distance), so it is also shown relative to the Earth: the lunar ellipse visibly *precesses*, a direct effect of the Sun's tidal perturbation that a two-body model would miss.

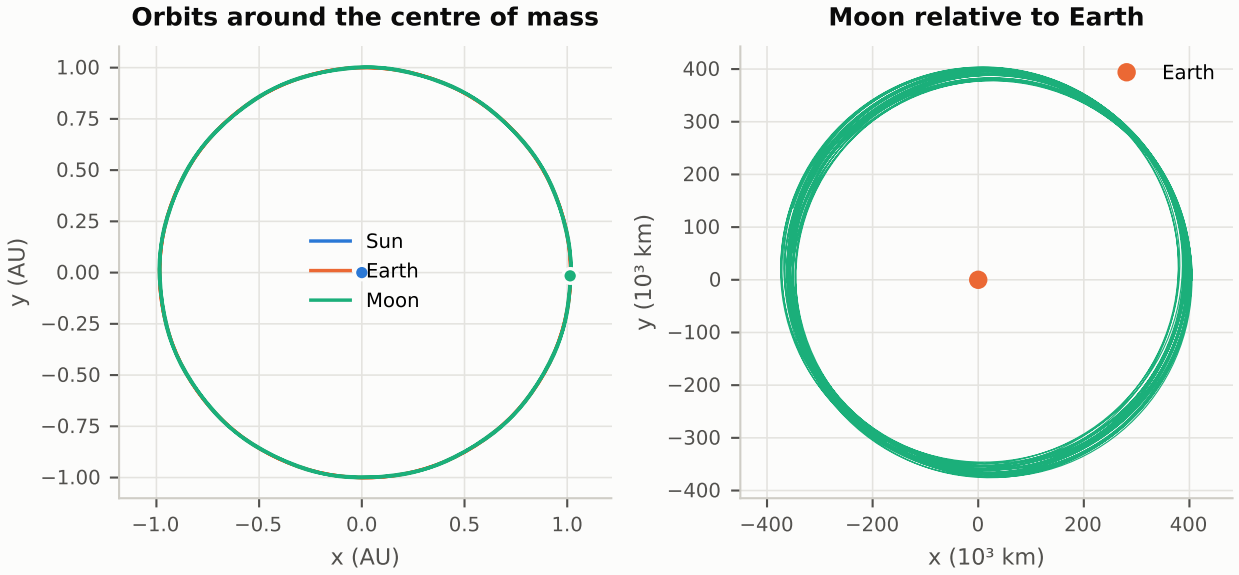



**Figure 1:** Sun–Earth–Moon system over two years. Left: orbits about the centre of mass (the Earth and the Moon overlap at this scale). Right: the Moon relative to the Earth; its orbit precesses under the Sun's perturbation.

The conservation laws are shown in Fig. 2. The energy error oscillates with the orbital period and stays below $6.41 \times 10^{-10}$: there is no secular drift, as expected from a symplectic method. The linear and angular momenta are conserved to $4.09 \times 10^{-15}$ and $1.67 \times 10^{-14}$ respectively, i.e. to machine precision; the slow growth is the accumulation of round-off errors. The time averages give $\langle K \rangle = 2.68 \times 10^{33}\,\mathrm{J}$ and $-\frac{1}{2}\langle U \rangle = 2.68 \times 10^{33}\,\mathrm{J}$, so $Q = 1.00002$: the virial theorem holds to better than $10^{-4}$.

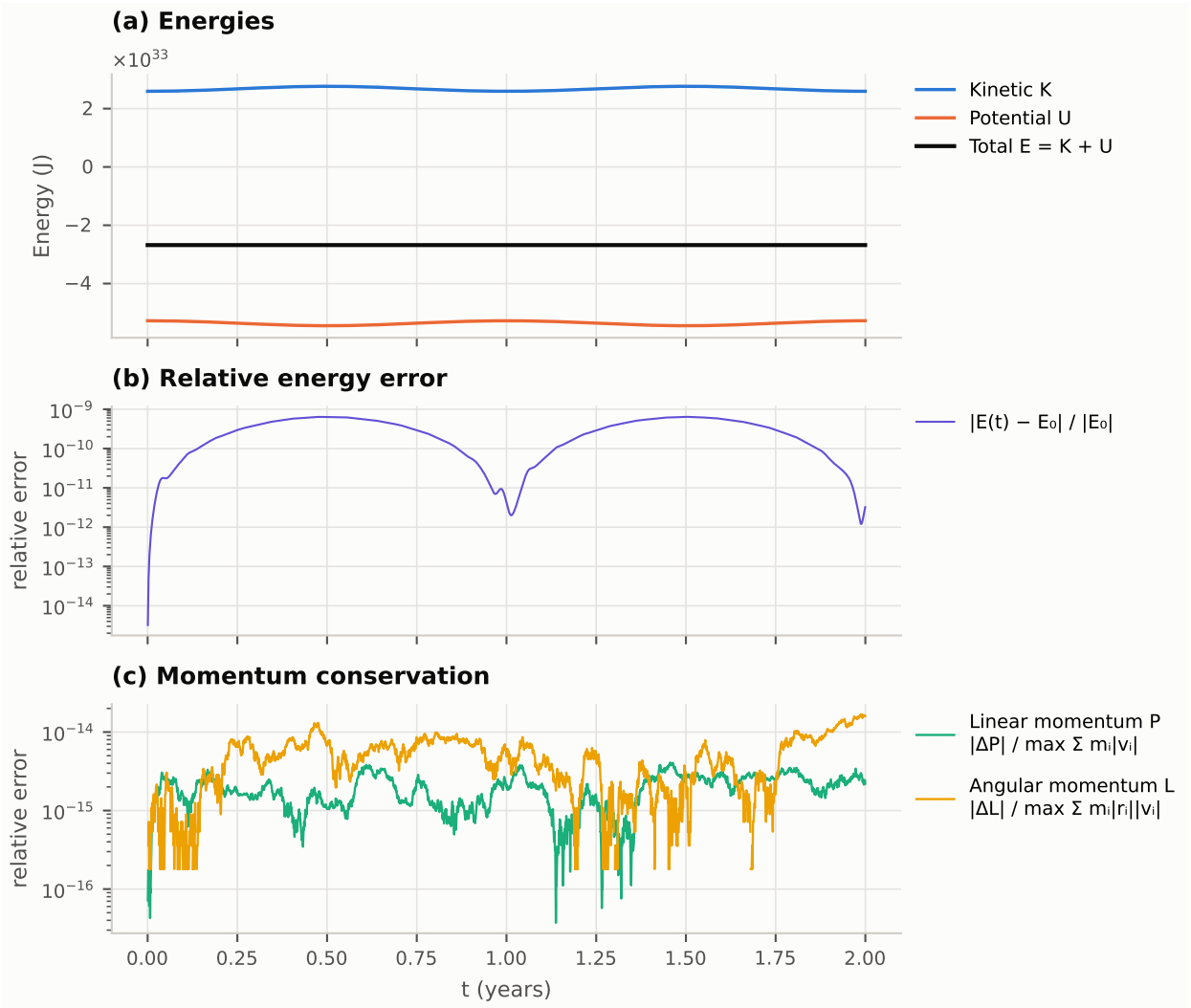



**Figure 2:** Conservation laws for the Sun–Earth–Moon run: (a) kinetic, potential and total energy; (b) relative energy error, bounded and oscillating; (c) relative errors of $\mathbf{P}$ and $\mathbf{L}$, at the level of round-off.

If the Sun is kept at rest initially (as in the original report) the total momentum is not zero and the CM moves at $V_{\mathrm{cm}} = 0.09$ m/s. Here this is negligible because the Sun dominates the mass, but in general it biases $\langle K \rangle$ and therefore $Q$.

## 4.2 Four bodies around a massive central body

This system reproduces Section 4.2 of the original report (Table 3): a central body of $2 \times 10^{40}$ kg (about $10^{10}$ solar masses, comparable to a galactic nucleus), two stars of about one solar mass in the $xy$ plane, and a light body on a polar orbit.

**Table 3:** Initial conditions of the four-body system ($d = 1.496 \times 10^{14}$ m, $M_\odot = 1.9891 \times 10^{30}$ kg). Integration time 15 years, $h = 1000$ s.

| | Body 1 | Body 2 | Body 3 | Body 4 |
|---|---|---|---|---|
| Mass (kg) | $2.02\,M_\odot$ | $0.98\,M_\odot$ | $2 \times 10^{40}$ | $2 \times 10^{20}$ |
| $\mathbf{r}_0$ (m) | $(0, 0, 0)$ | $(d, 0, 0)$ | $(d/2, 0, 0)$ | $(d/2, 0, 10^{14})$ |
| $\mathbf{v}_0$ | $(0, -\sqrt{Gm_3/d}, 0)$ | $(0, \sqrt{Gm_3/d}, 0)$ | $\mathbf{0}$ | $(-\sqrt{Gm_3/10^{14}}, 0, 0)$ |

The original report intended circular orbits, but the speeds use the distance $d$, which is *twice* the actual distance $r = d/2$ to the central body. Then $v^2 = GM/(2r)$ and the vis-viva equation gives $a = 2r/3$: the orbits are ellipses with eccentricity $e = 0.5$ whose apocentre is the starting point. The simulation measures $e = 0.5000$, confirming the analysis. Body 4 starts at the correct distance and moves on a circle.

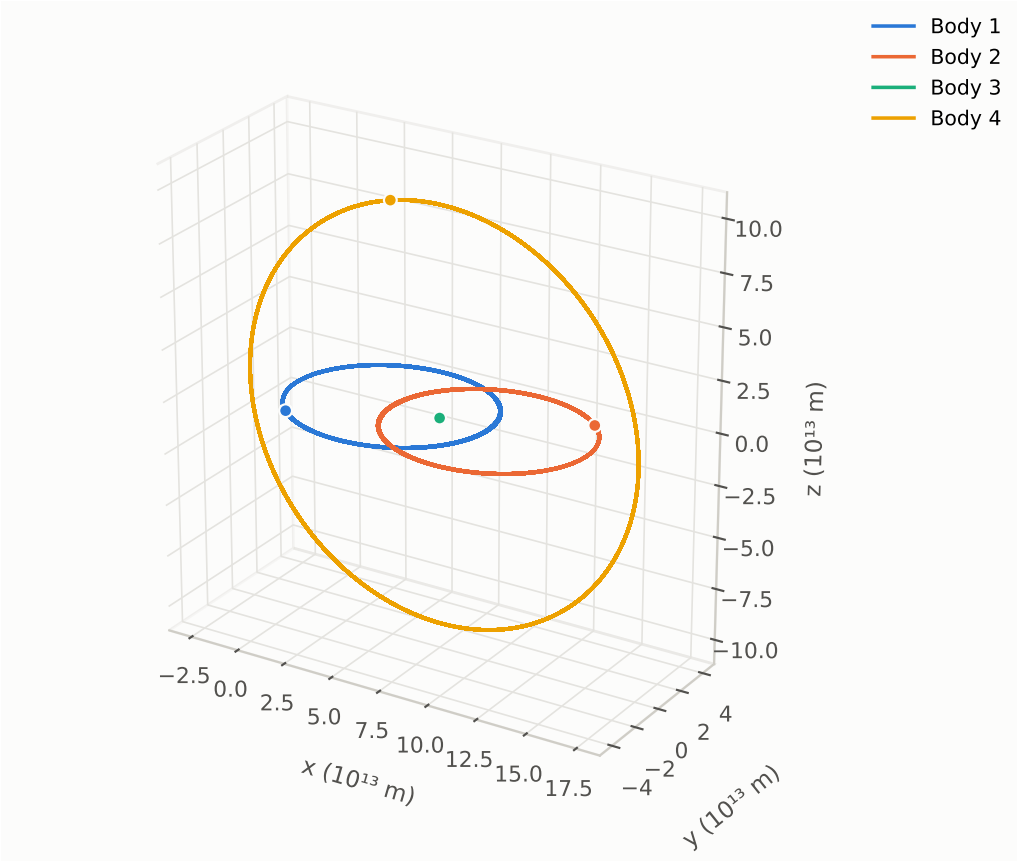



**Figure 3:** Orbits of the four-body system over 15 years (about 250 orbits of bodies 1 and 2). The ellipses are closed and stable; the massive body barely moves.

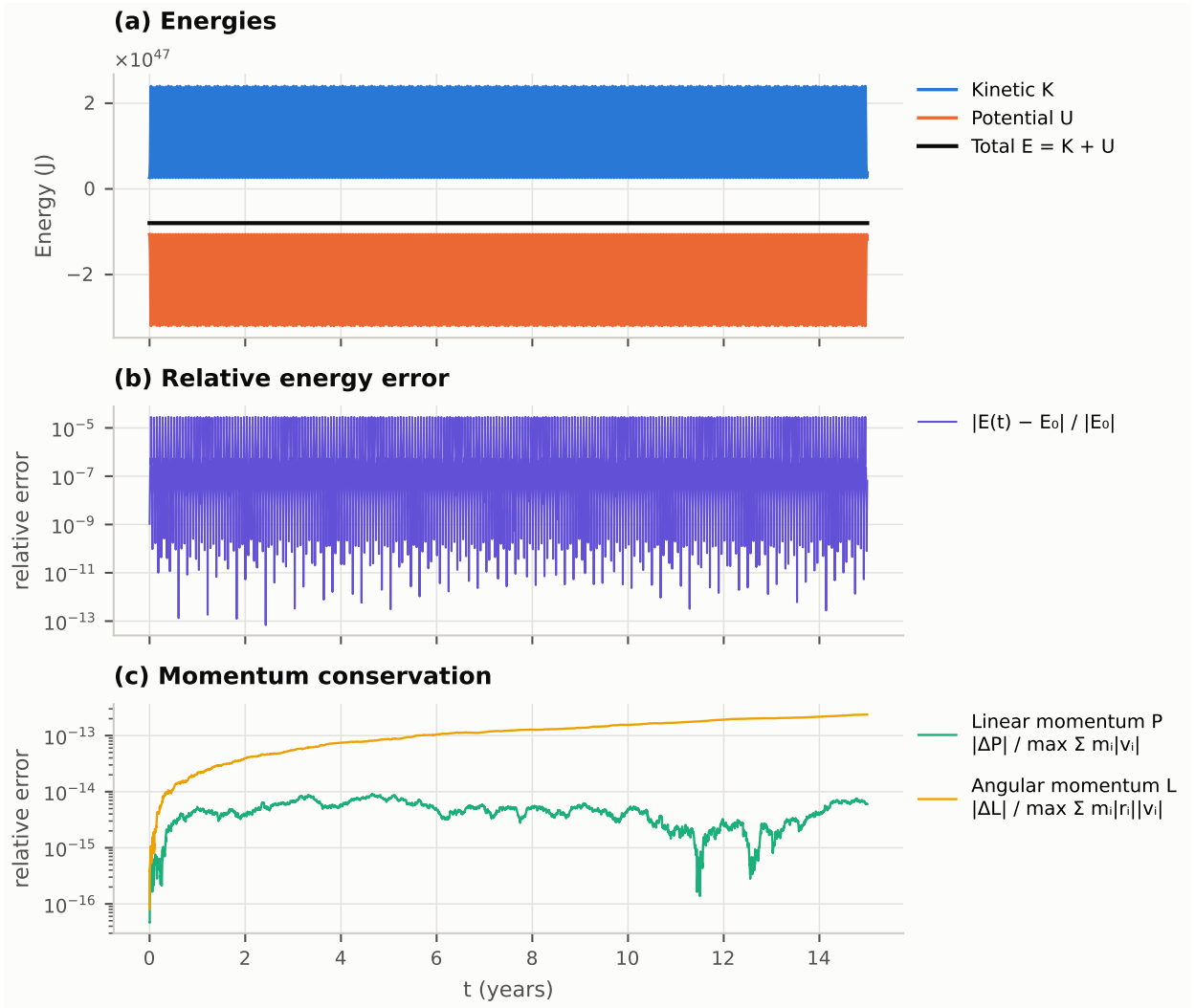



**Figure 4:** Conservation laws for the four-body system. The energy error peaks at pericentre passages, where the time step is relatively largest, but does not grow.

With eccentric orbits the energy error peaks at each pericentre passage, where the orbital time scale is shortest, and reaches $2.87 \times 10^{-5}$ (Fig. 4); yet the envelope is flat over 15 years. $\mathbf{P}$ and $\mathbf{L}$ are conserved to $9.05 \times 10^{-15}$ and $2.37 \times 10^{-13}$, and the virial ratio is $Q = 0.99974$.

### 4.3 Chaos in a six-body system

Six bodies of $1 \times 10^{10}\,\mathrm{kg}$ start on the $\pm x$, $\pm y$, $\pm z$ semi-axes at $1 \times 10^{5}\,\mathrm{m}$, each with the speed ($\approx 2.1\,\mathrm{mm/s}$) of a circular orbit around the next one (Table 4). We integrate four versions for 10 years: the reference and three runs in which a single quantity is changed by 0.1 %.

**Table 4:** Six-body system ($u = 1.5 \times 10^{-3}\,\mathrm{m/s}$). Integration time 10 years, $h = 1000\,\mathrm{s}$.

| | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $\mathbf{r}_0/10^5\,\mathrm{m}$ | $(1,0,0)$ | $(0,1,0)$ | $(0,0,1)$ | $(-1,0,0)$ | $(0,-1,0)$ | $(0,0,-1)$ |
| $\mathbf{v}_0/u$ | $(1,1,0)$ | $(0,1,1)$ | $(-1,0,1)$ | $(-1,-1,0)$ | $(0,-1,-1)$ | $(1,0,-1)$ |

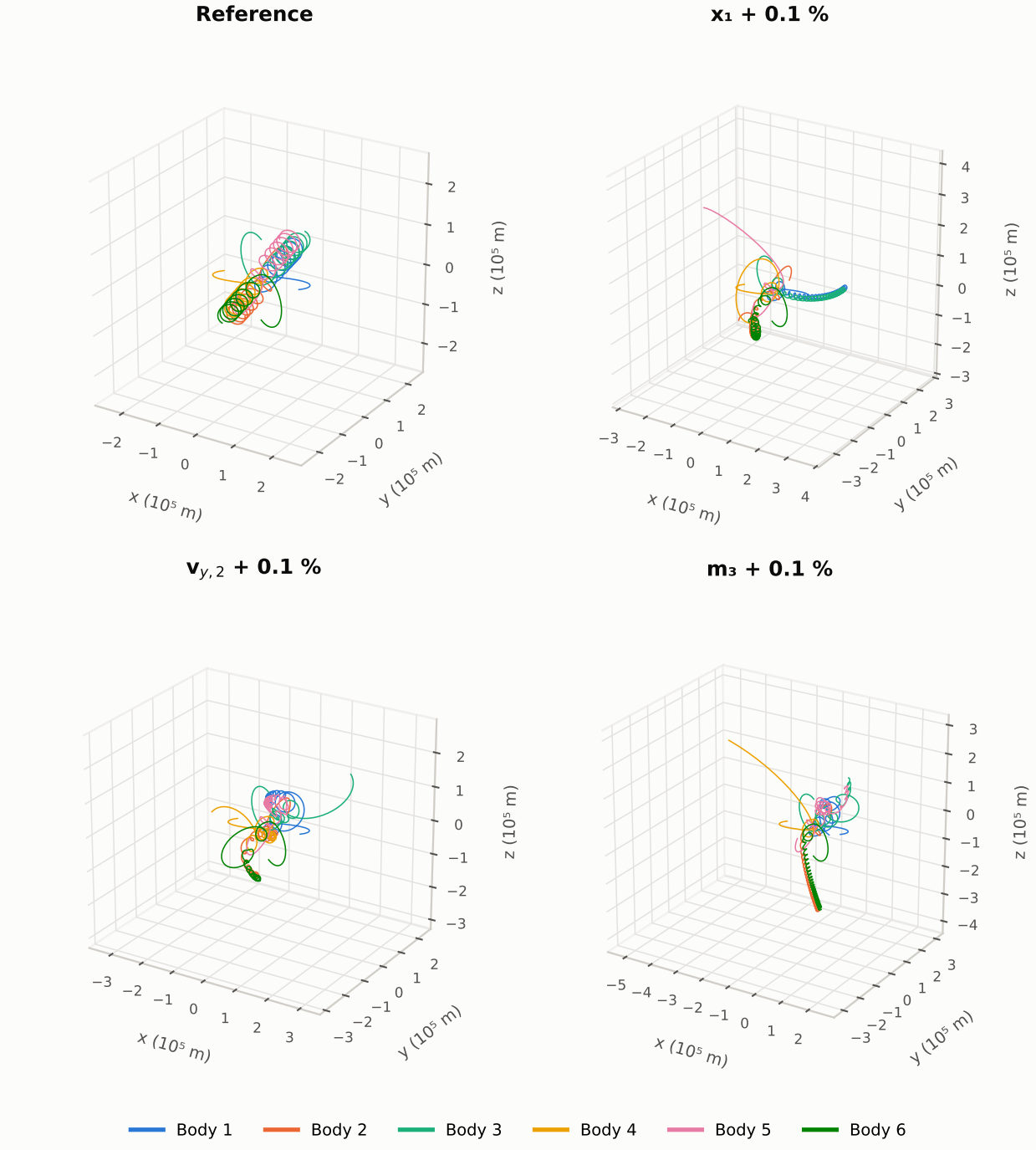



**Figure 5:** Six-body system: reference run and three runs with a 0.1 % change in $x_1$, $v_{y,2}$ and $m_3$. Bodies pair up into binaries and some are ejected, differently in each run.

Figure 5 shows that the final configurations are qualitatively different. Figure 6 quantifies this: the separation from the reference grows exponentially (a straight line on a log scale) after the first close encounter, around $t \approx 2$ years, until it saturates at the size of the system.

The most instructive curve is the last one: the *same* initial condition integrated with $h$ and with $h/2$ also diverges exponentially, reaching 835 m after 10 years. The truncation error acts as one more perturbation, so beyond a few Lyapunov times neither run reproduces the exact trajectory. This does not invalidate the simulation: $E$, $\mathbf{P}$ and $\mathbf{L}$ are still conserved ($6.68 \times 10^{-7}$, $3.34 \times 10^{-15}$, $7.14 \times 10^{-15}$), and the shadowing lemma guarantees that the numerical orbit stays close to *some* exact orbit with slightly different initial conditions. Statistical properties are reliable; individual trajectories are not.

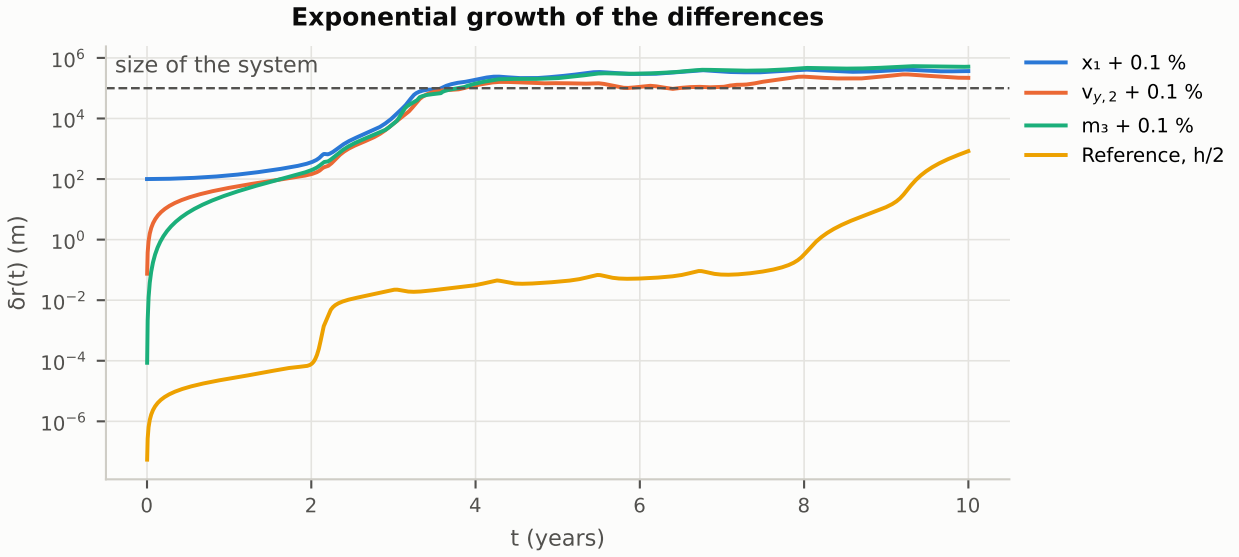



**Figure 6:** Separation $\delta r(t)$ between each perturbed run and the reference. The yellow curve compares the same initial condition integrated with $h$ and $h/2$.

Benettin's method (Fig. 7) gives $\lambda_{\max} \approx 3.37\ \mathrm{yr}^{-1}$ after 10 years, i.e. a Lyapunov time of about 108 days. The estimate jumps after the first close encounter and is still drifting slowly, a sign that the chaotic phase starts at $t \approx 2$ yr. Over ten years an initial error is amplified by roughly 15 orders of magnitude: not even double-precision initial conditions (relative error $10^{-16}$) would allow a deterministic prediction.

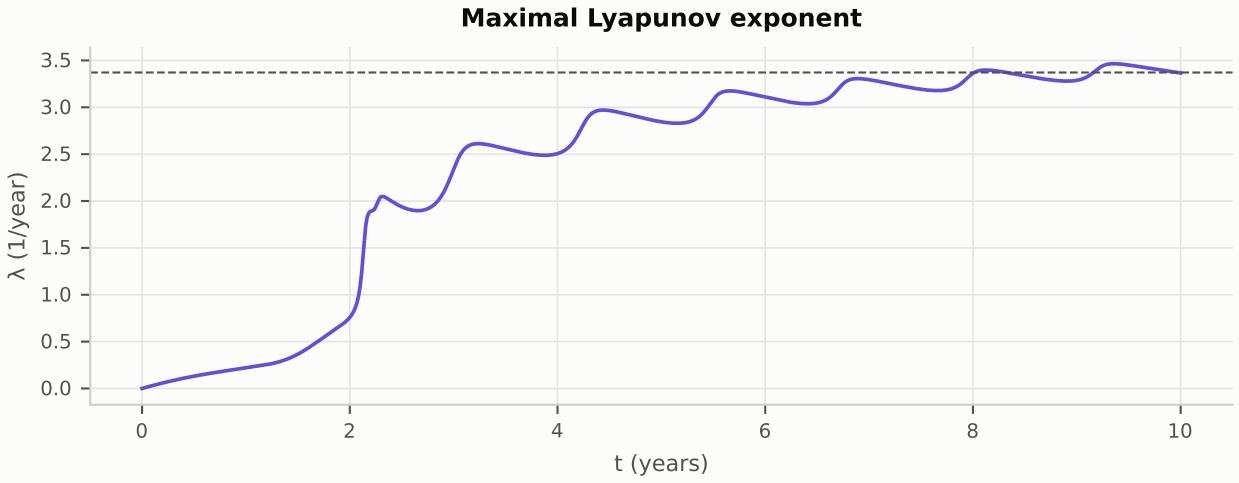



**Figure 7:** Running estimate of the maximal Lyapunov exponent (Benettin's method, renormalization every 20 steps, $\delta_0 = 10^{-8}$). The dashed line is the average over the last 20 % of the run.

## 4.4 Comparison of integrators

To isolate the effect of the integrator we use the figure-eight orbit of Section 4.5, with $G = m = 1$ and period $T \approx 6.326$. Figure 8 shows the maximum energy error per period over 200 periods with a deliberately coarse step, $h = T/300$. Explicit Euler gains energy at every step and the system disintegrates within ten periods. RK4 starts with a tiny error but drifts *linearly*. The three symplectic schemes show no drift at all, as the shadow-Hamiltonian argument predicts; at equal cost Verlet is far more robust than its formal order suggests, and Yoshida-4 combines high order with bounded error.

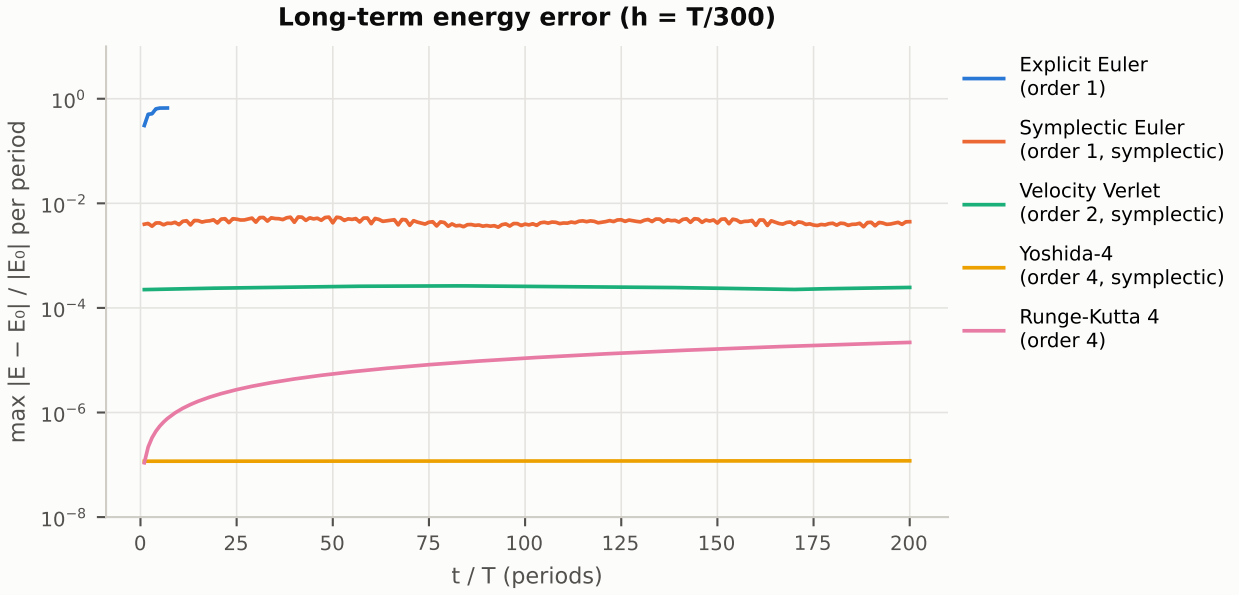



**Figure 8:** Long-term energy error (maximum per period) for $h = T/300$ on the figure-eight orbit. Only the non-symplectic methods show a secular drift.

Figure 9 plots the position error after one period against the number of force evaluations, the real cost. The fitted slopes give orders 0.93, 0.99, 2.00, 4.00 and 4.13, matching the theoretical values 1, 1, 2, 4 and 4.

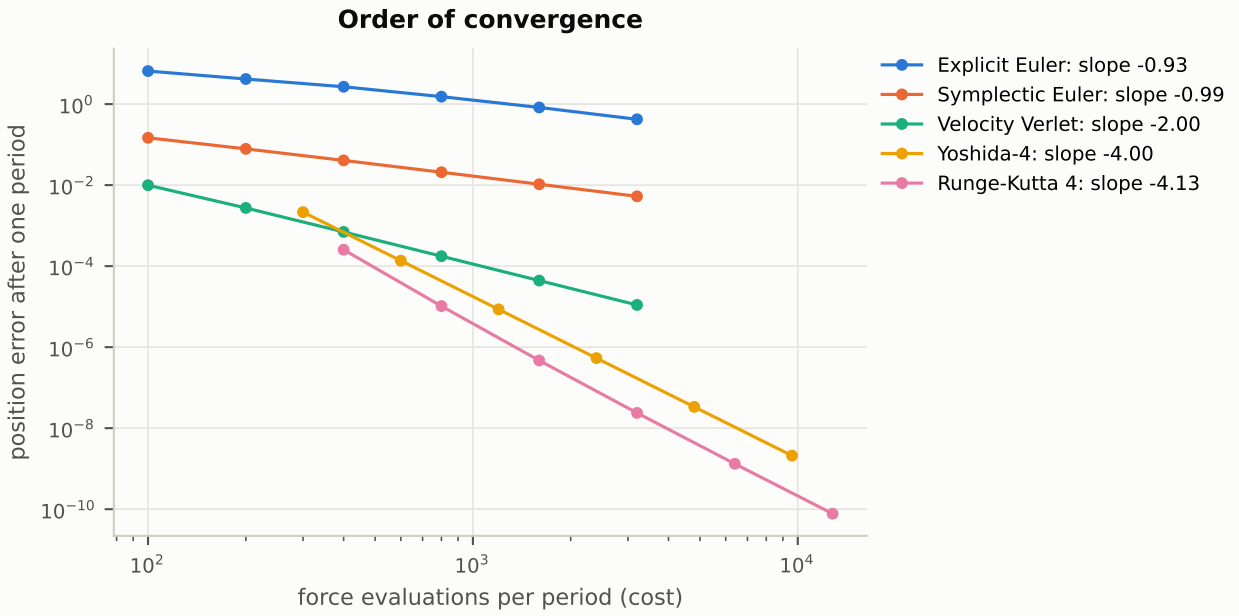



**Figure 9:** Order of convergence: position error after one period against the number of force evaluations. Slopes are fitted on the three smallest steps.

## 4.5 The figure-eight choreography

Moore (1993) found numerically, and Chenciner and Montgomery (2000) proved, that three equal masses can chase each other along a single figure-eight curve, a third of a period apart. The initial conditions are $\mathbf{r}_1 = -\mathbf{r}_2 = (-0.97000436, 0.24308753)$, $\mathbf{r}_3 = \mathbf{0}$, $\mathbf{v}_3 = (-0.93240737, -0.86473146)$, $\mathbf{v}_1 = \mathbf{v}_2 = -\mathbf{v}_3/2$. The total angular momentum is exactly zero, even though the bodies rotate: the two lobes are travelled in opposite senses.

With $h = T/6000$ (Fig. 10) the bodies return to their initial positions within $3.17 \times 10^{-6}$ after one period and $5.03 \times 10^{-5}$ after ten (the initial data themselves are given to eight digits). The energy error is bounded at $6.55 \times 10^{-7}$ and $Q = 1.00000$.

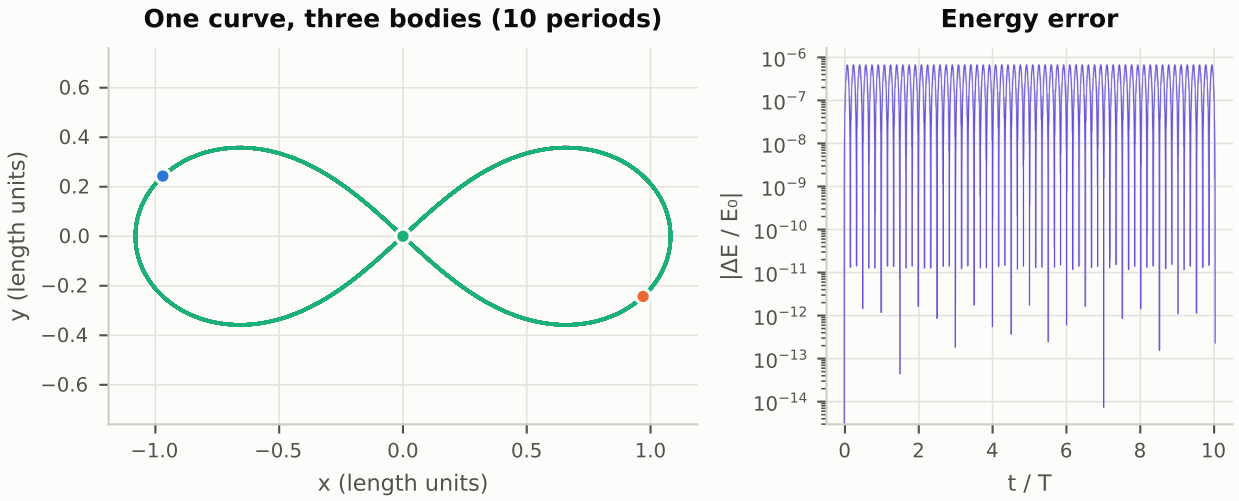



**Figure 10:** Figure-eight choreography over ten periods: the three orbits coincide. Right: bounded relative energy error.

Changing one velocity component by 0.1 % produces a separation that grows only slowly and *without* the exponential trend of the six-body system (Fig. 11), reaching 0.040 after ten periods. The orbit is surrounded by invariant KAM tori: it is stable.

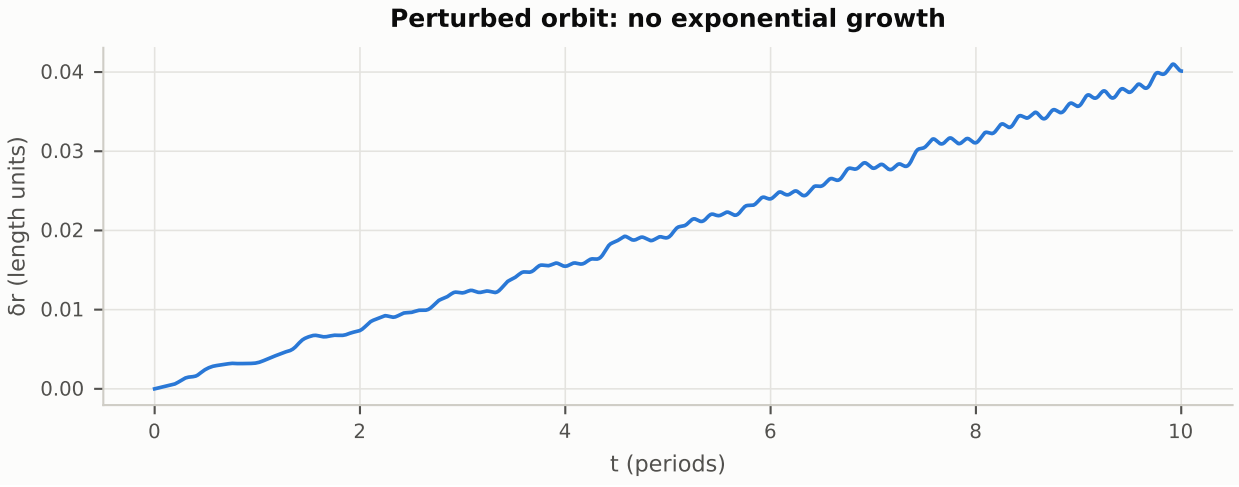



**Figure 11:** Separation between the figure-eight orbit and a copy with $v_{x,1}$ changed by 0.1 %: slow, non-exponential growth.

## 4.6 Stability of Lagrange's triangle

Lagrange (1772) showed that three bodies at the vertices of an equilateral triangle of side $d$, rotating rigidly with $\omega^2 = GM/d^3$, form an exact solution for any masses. Linearizing in the rotating frame, the perturbation exponents satisfy

$$\lambda^4 + \omega^2\lambda^2 + \tfrac{27}{4}\,\beta\,\omega^4 = 0, \qquad \beta = \frac{m_1m_2 + m_2m_3 + m_3m_1}{M^2}. \tag{12}$$

The roots are purely imaginary (stable oscillations) only if $27\beta < 1$, which is Routh's criterion. For the Sun, Jupiter and a Trojan asteroid ($\beta \approx 10^{-3}$) the configuration is stable, which is why

Trojan asteroids cluster at Jupiter's L4 and L5 points. For equal masses $\beta = 1/3$, the roots satisfy $\lambda^2 = \omega^2(-1 \pm i\sqrt{8})/2$ and the growth rate is $\mathrm{Re}\,\lambda = \omega/\sqrt{2}$.

In the simulation no perturbation is added by hand: the truncation error of Verlet is enough (Fig. 12). For equal masses the deformation of the triangle grows exponentially from $10^{-6}$ to order one, and a linear fit of $\ln\delta$ gives $\sigma = 1.225$, against the theoretical 1.225. For the Sun–Jupiter–Trojan masses the deformation stays below $5.01 \times 10^{-7}$. The small periodic deformation at the start is not an instability: Verlet solves exactly a shadow Hamiltonian whose circular orbit has a slightly different radius.

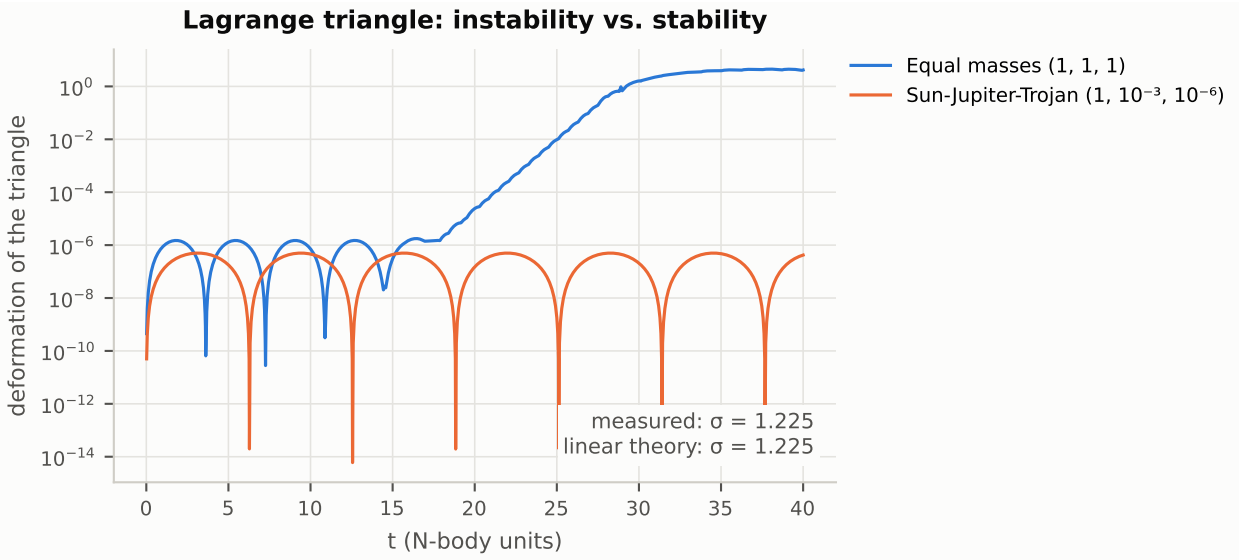



**Figure 12:** Deformation of Lagrange's triangle (largest relative change of a side). Equal masses: exponential growth at the rate predicted by linear theory. Sun–Jupiter–Trojan: bounded.

## 4.7 The Solar System and Kepler's third law

The Sun and the eight planets are initialized with their real masses, semi-major axes, eccentricities and inclinations (Table 5). Each planet starts at perihelion with speed $v_p = \sqrt{GM_\odot(1+e)/(a(1-e))}$ and a random orbital phase; its orbital plane is tilted about the Sun–perihelion line. These are illustrative initial conditions, not the ephemeris of a given date. The step $h = 0.5$ days is set by Mercury (176 steps per orbit), and the run covers one orbit of Neptune, 165 years.

**Table 5:** Planetary parameters and orbital periods measured in the simulation (counting revolutions of the polar angle about the Sun).

| Planet | Mass (kg) | $a$ (AU) | $e$ | $i$ (°) | $T$ measured (yr) |
|---|---|---|---|---|---|
| Mercury | $3.3011 \times 10^{23}$ | 0.387 | 0.206 | 7.00 | 0.2411 |
| Venus | $4.8675 \times 10^{24}$ | 0.723 | 0.007 | 3.39 | 0.6152 |
| Earth | $5.9722 \times 10^{24}$ | 1.000 | 0.017 | 0.00 | 1 |
| Mars | $6.4171 \times 10^{23}$ | 1.524 | 0.093 | 1.85 | 1.881 |
| Jupiter | $1.8982 \times 10^{27}$ | 5.203 | 0.049 | 1.30 | 11.85 |
| Saturn | $5.6834 \times 10^{26}$ | 9.555 | 0.057 | 2.49 | 29.36 |
| Uranus | $8.6810 \times 10^{25}$ | 19.22 | 0.046 | 0.77 | 84.91 |
| Neptune | $1.02413 \times 10^{26}$ | 30.11 | 0.011 | 1.77 | 163.4 |

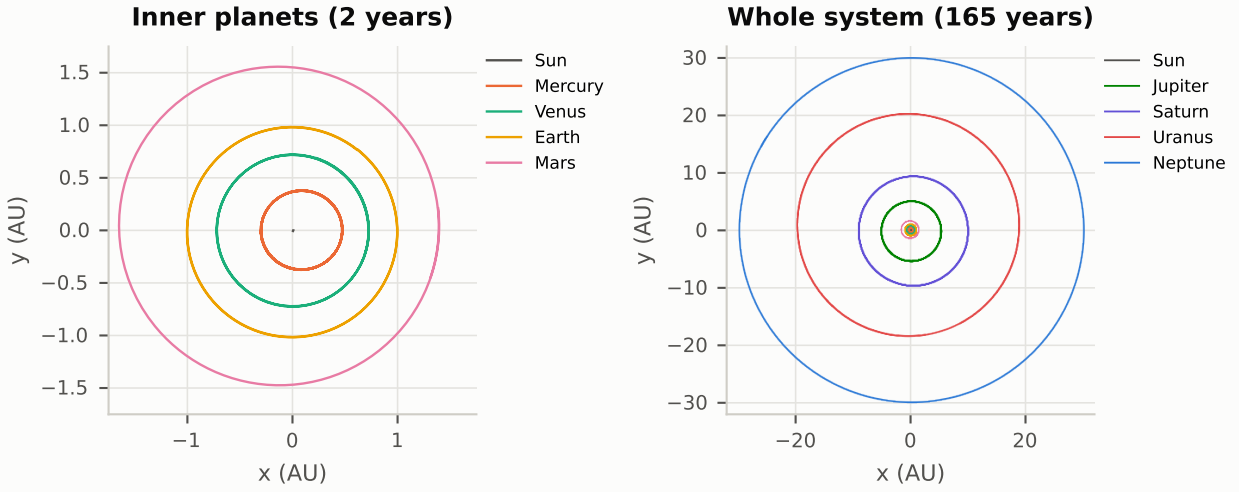



**Figure 13:** Solar System: inner planets over two years (left) and the whole system over one orbit of Neptune (right).

A log-log fit of the measured periods against the semi-major axes (Fig. 14) gives a slope of 1.4989, compared with 3/2 from Kepler's third law. The small deviations of the outer planets are real: Jupiter and Saturn perturb each other and their neighbours, and for Uranus and Neptune a single, eccentric orbit is too short to average the angular speed. Over 165 years the energy is conserved to $7.22 \times 10^{-7}$, the momenta to $1.27 \times 10^{-14}$ and $1.19 \times 10^{-14}$, and $Q = 0.99967$.

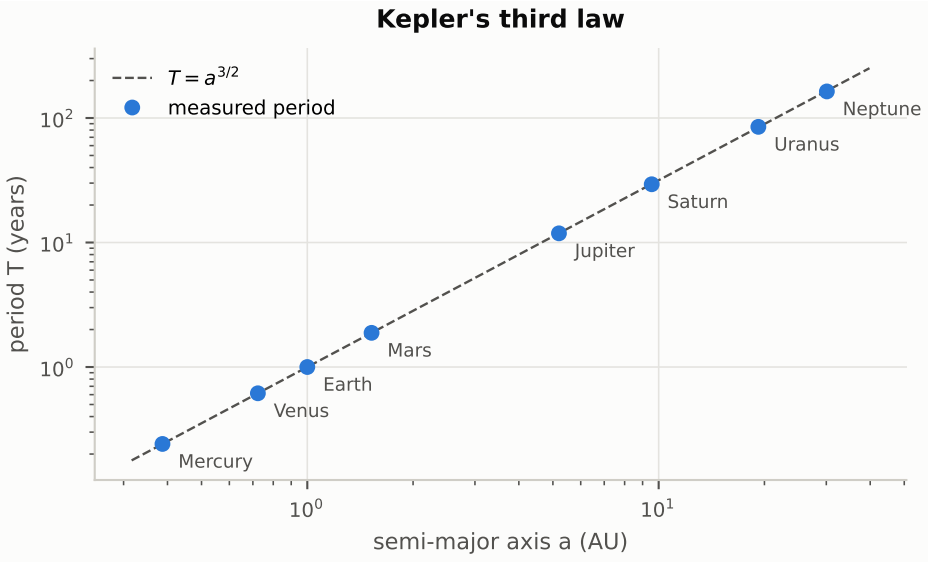



**Figure 14:** Kepler's third law: orbital periods measured in the simulation against the semi-major axis, with the line $T = a^{3/2}$ (years and AU).

## 4.8 Star clusters and the virial theorem

Finally we follow 400 equal-mass "stars" in $N$-body units with softening $\varepsilon = 0.05$ and $h = 0.002$. A *Plummer sphere*, sampled from its equilibrium distribution function, is born virialized and stays near $Q = 1$ (late-time average 1.01), fluctuating because $N$ is finite.

A *cold collapse* starts from a homogeneous sphere with every star at rest ($Q = 0$). Gravity makes it collapse in about one free-fall time $t_{ff} = \frac{\pi}{2}\sqrt{R^3/2GM} \approx 1.11$; at maximum compression $Q$ overshoots to 1.72, the cluster rebounds, and within a few $t_{ff}$ it settles into a quasi-stationary state with $\langle Q \rangle = 1.03$ (Fig. 15). This *violent relaxation* (Lynden-Bell, 1967) happens on a dynamical

time scale, without star–star collisions, and is believed to shape elliptical galaxies. The Lagrangian radii show a dense core forming while the outer 10 % of the mass expands into a halo, part of which escapes (Fig. 16). The softened energy is conserved to $1.19 \times 10^{-3}$ despite the violent compression.

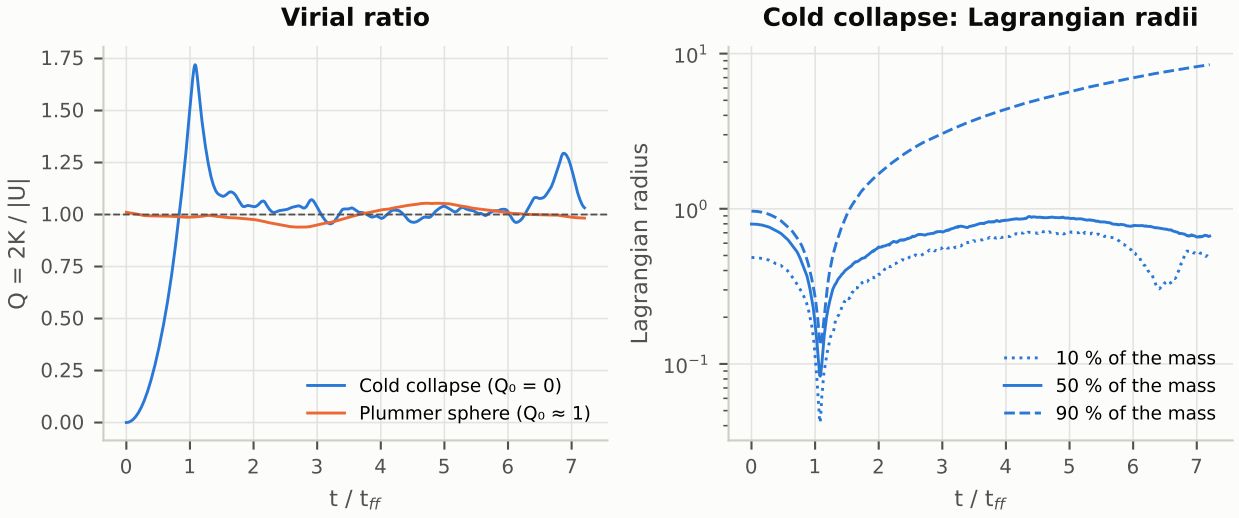



**Figure 15:** Left: virial ratio $Q = 2K/|U|$ for the cold collapse and the Plummer sphere. Right: Lagrangian radii containing 10, 50 and 90 % of the mass during the cold collapse.

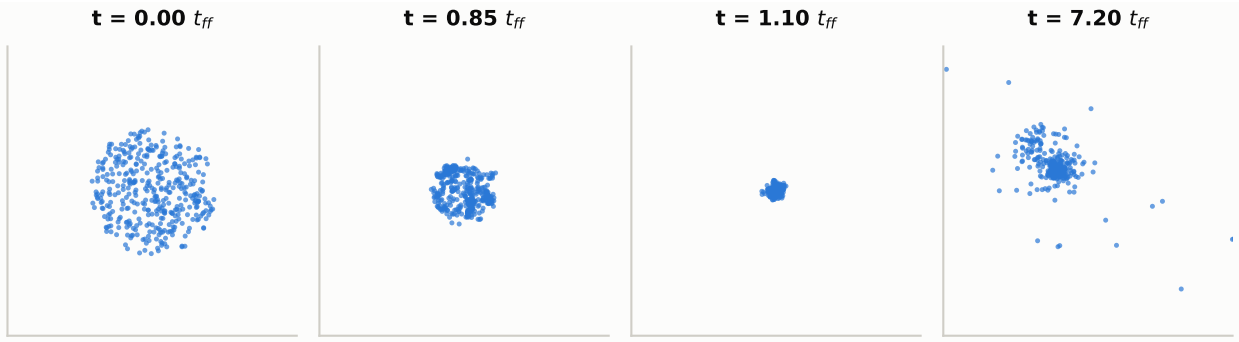



**Figure 16:** Snapshots of the cold collapse ($xy$ projection): initial sphere, contraction, maximum compression and the final relaxed cluster with escaping stars.

## 4.9 Videos

The repository also contains 3D videos (MP4) of these experiments, rendered with a rotating camera and fading trails (`videos/`). Figure 17 shows four frames.

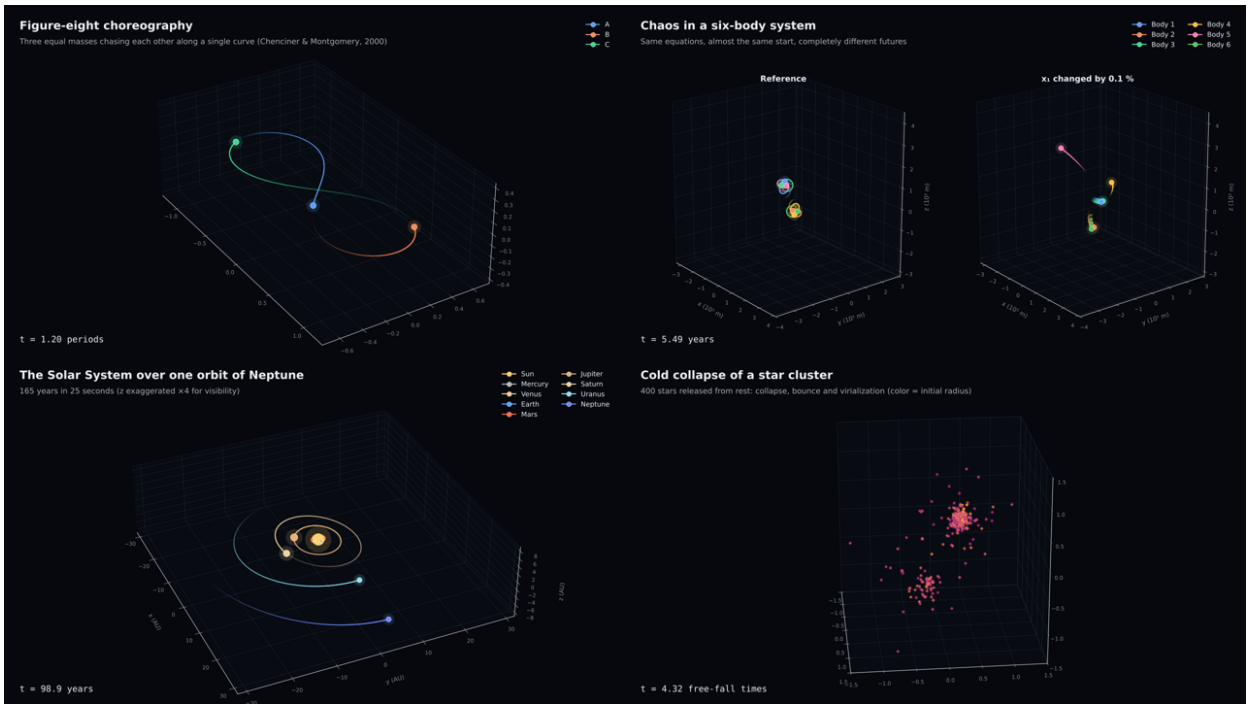



**Figure 17:** Frames from the videos: figure-eight choreography, chaos in the six-body system, the Solar System over one orbit of Neptune and the cold collapse of a star cluster.

## 5 Discussion

Table 6 collects the conservation errors of all runs. Linear and angular momentum are conserved to machine precision in every case, as expected from a method built on antisymmetric central pair forces. The energy error is always bounded and its size is controlled by the ratio of the time step to the shortest dynamical time: it is largest for the eccentric orbits of the four-body system and for the dense core of the collapsing cluster.

**Table 6:** Maximum conservation errors and virial ratio for every run (velocity Verlet).

| Run | $\lvert\Delta E/E_0\rvert$ | $\lvert\Delta\mathbf{P}\rvert$ (rel.) | $\lvert\Delta\mathbf{L}\rvert$ (rel.) | $Q$ |
|---|---|---|---|---|
| Sun–Earth–Moon (2 yr) | $6.41 \times 10^{-10}$ | $4.09 \times 10^{-15}$ | $1.67 \times 10^{-14}$ | 1.00002 |
| Four bodies (15 yr) | $2.87 \times 10^{-5}$ | $9.05 \times 10^{-15}$ | $2.37 \times 10^{-13}$ | 0.99974 |
| Six bodies (10 yr) | $6.68 \times 10^{-7}$ | $3.34 \times 10^{-15}$ | $7.14 \times 10^{-15}$ | — |
| Figure-eight (10 $T$) | $6.55 \times 10^{-7}$ | — | — | 1.00000 |
| Solar System (165 yr) | $7.22 \times 10^{-7}$ | $1.27 \times 10^{-14}$ | $1.19 \times 10^{-14}$ | 0.99967 |

The virial theorem is satisfied to $10^{-4}$ or better for all bound, bounded systems. It does not apply to the six-body system, where bodies escape, nor at early times of the cold collapse, which is precisely what makes the latter a good illustration of relaxation towards virial equilibrium.

Compared with the original 2023 work, three corrections are worth noting. First, the Moon's mass in Table 1 of that report ($2 \times 10^{40}$ kg) was a typo for $7.342 \times 10^{22}$ kg. Second, the Earth's orbit has eccentricity close to zero, not to one. Third, the orbits of the four-body system are not circular but have $e = 0.5$, as explained by the vis-viva equation. In addition, measuring errors relative to the initial value (rather than subtracting the mean) reveals the slow round-off drift of $\mathbf{L}$ that the mean hides.

The main limitations are those of any fixed-step direct-summation code. Close encounters

(for example Burrau's Pythagorean three-body problem) require either adaptive time steps, which must be designed carefully to preserve symplecticity, or regularization. The $\mathcal{O}(N^2)$ cost limits the method to a few thousand bodies; tree codes ($\mathcal{O}(N \log N)$) or fast multipole methods are needed beyond that.

## 6 Conclusions

- The velocity Verlet method conserves linear and angular momentum to machine precision and keeps the energy error bounded over thousands of orbits, because it is symplectic. Non-symplectic methods, even of higher order, drift.
- The measured orders of convergence (2.00 for Verlet, 4.00 for Yoshida-4) agree with theory, and Yoshida-4 offers the best accuracy per cost while remaining symplectic.
- Analytical results are reproduced quantitatively: Kepler's third law (slope 1.4989), the eccentricity predicted by the vis-viva equation (0.5000), the linear instability rate of Lagrange's triangle (1.225 vs. 1.225), and the virial theorem.
- The $N$-body problem is chaotic in general: the six-body system has a Lyapunov time of about 108 days, and even the integration step changes the trajectory after a few Lyapunov times. Long-term predictions are only meaningful in a statistical sense, which is guaranteed by the conservation laws and the shadowing property.

## A Reproducing the results

```
git clone https://github.com/jesus-garcia-izquierdo/N-body-problem.git
cd N-body-problem
pip install -e .
python examples/01_sun_earth_moon.py      # ... up to 08_cluster_virial.py
python tools/build_documents.py           # regenerates this report
```

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